

SIMATS ENGINEERING
CO3 - ASSESSMENT TOOL - 2: Case Study

COURSE NAME: Cloud Computing and Big Data Analytics **COURSE CODE:** CSA1522

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COURSE OUTCOME COVERED

CO3: *Implement cloud-based solutions using cloud storage, web APIs, and platform services for real-world applications. (BL3)*

Dave's Taxonomy: Precision (Level 3)
Question Count: 3

Total Marks: 30

Code of Conduct

I BALA S (192524427) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: bala

Assessment Tasks - Case Study

1. Case Study 1: Smart Retail Inventory Management System

A retail chain implements a cloud-based inventory system connected to store devices. Clients (POS systems and dashboards) communicate via APIs to centralized cloud services. Cloud storage stores product data, transaction logs, and analytics datasets. The system uses platform services for analytics and demand forecasting. Secure access is maintained through VPNs, firewalls, and API gateways. This solution integrates infrastructure, services, and web applications for efficient operations.

2. Case Study 2: Cloud-Based Document Collaboration System

An enterprise deploys a document collaboration tool similar to Google Docs. Users access the platform via browsers, enabling real-time editing and sharing. Documents are stored in distributed cloud storage with version control mechanisms. Web APIs enable synchronization across multiple clients and devices. Network infrastructure ensures low latency and high availability globally. Security includes encryption, identity management, and access permissions for collaboration.

3. Case Study 3: Online Food Ordering Platform

A startup builds a cloud-based food ordering system accessible via web browsers and mobile clients. Frontend clients interact with backend services through RESTful Web APIs hosted on a cloud platform. Cloud storage is used to store menu images, invoices, and customer data securely. The infrastructure uses virtual machines and managed Kubernetes for scalability and reliability. Security is enforced

through authentication tokens, HTTPS, and role-based access control. The system demonstrates integration of clients, APIs, storage, and platform services in real time.

Case Study Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Understanding of Case	25%	Clearly identifies all technical and business requirements	Identifies most requirements	Basic understanding only	Poor understanding of case
Application of Concepts	25%	Applies suitable cloud concepts and tools effectively	Good application with minor issues	Partial application	Weak or incorrect application
Analysis	20%	Strong analysis with clear trade-off reasoning	Reasonable analysis	Limited analysis	Minimal analysis
Solution Design	20%	Solution is feasible, practical, and well justified	Reasonable solution with minor gaps	Basic solution	Unclear solution
Clarity	10%	Well-structured and easy to follow	Mostly clear	Somewhat unclear	Poorly organized

Case Study 1: Smart Retail Inventory Management System

- A retail chain with multiple branches wants to manage its inventory efficiently using a cloud-based inventory management system.
- Every store has Point of Sale (POS) systems, barcode scanners, and web dashboards that act as clients.
- These clients communicate with centralized cloud services through REST APIs.
- The cloud system stores product details, sales transactions, customer orders, supplier information, and inventory records in cloud storage.
- Since all stores use the same centralized database, inventory is updated instantly whenever a sale or restocking occurs.

Functional Requirements

- Real-time inventory tracking.
- Product and supplier management.
- Sales and billing through POS.

- Automatic inventory updates.
- Order and stock replenishment management.
- Sales reporting and analytics. .

Non-Functional Requirements

- High availability (24×7).
- Scalability for adding more stores.
- Data security and privacy.
- Fast response time.

Infrastructure as a Service (IaaS)

- Virtual machines host inventory applications.
- Cloud networking connects all stores.
- Storage volumes keep inventory databases.

Platform as a Service (PaaS)

- Machine learning predicts future demand.
- Analytics services generate sales reports.
- Developers deploy applications without managing servers.

Software as a Service (SaaS)

- Store managers access inventory through web browsers.
- Dashboards display stock levels and reports.

Working Process

Step 1: Customer purchases a product.

Step 2: POS records the transaction.

Step 3: Transaction is securely sent to the cloud through REST APIs.

Step 4: Inventory service updates the stock immediately.

Step 5: Transaction details are stored in cloud storage.

Step 6: Analytics service processes the sales data.

Step 7: Demand forecasting predicts future inventory needs.

Step 8: Managers monitor stock using cloud dashboards.

Benefits

- Real-time inventory updates.
- Better stock management.
- Faster order processing.
- Improved sales forecasting.

Case Study 2: Cloud-Based Document Collaboration System

- A cloud-based document collaboration system allows multiple users to create, edit, and share documents in real time through a web browser.
- Similar to Google Docs, the application stores documents in distributed cloud storage and synchronizes changes across all connected devices using Web APIs.

LAYERS:

- Client Layer: Web browsers and mobile applications used by users.
- Application Layer: Cloud servers handle authentication, document editing
- API Layer: Web APIs synchronize document updates between multiple users
- Storage Layer: Distributed cloud storage stores documents with automatic replication and version control.
- Network Layer: Global Content Delivery Network (CDN) and load balancers provide low latency
- Security Layer: Encryption, identity management, authentication, and role-based access control protect documents.

Process:

1. Users log in through a browser or mobile application.
2. Documents are created or opened from cloud storage.
3. Multiple users edit the same document simultaneously.
4. Web APIs instantly synchronize changes across all connected devices.
5. Version control stores document history and allows recovery of previous versions.
6. Security mechanisms verify user identity and enforce access permissions.

Cloud Features

- Real-time collaboration
- Distributed cloud storage
- Automatic synchronization
- Version control
- Auto-scaling and load balancing

Security Measures

- Data encryption (in transit and at rest)
- Identity and Access Management (IAM)
- Multi-Factor Authentication (MFA)
- Role-Based Access Control (RBAC)

Benefits

- Real-time teamwork and productivity
- Access documents from anywhere
- Automatic backup and recovery
- Reduced infrastructure maintenance

Case Study 3: Online Food Ordering Platform

- A startup develops a cloud-based food ordering platform that customers can access through web browsers and mobile apps.
- The frontend communicates with backend services using RESTful Web APIs hosted on a cloud platform.
- Cloud storage securely stores menu images, invoices, and customer data.
- Virtual Machines (VMs) and managed Kubernetes ensure scalability, reliability, and high availability.

Client Applications (Web & Mobile):

- Customers access the platform through web browsers or mobile apps.
- The frontend provides an easy-to-use interface for browsing restaurants, selecting food items, placing orders, and tracking deliveries.

RESTful Web APIs:

- The frontend communicates with backend services using RESTful Web APIs.
- These APIs process user requests such as login, menu retrieval, order placement, payment, and order tracking.

Cloud Storage:

- Cloud storage securely stores menu images, customer information, invoices, and order history.
- It provides high availability, backup, and easy access to data.

Virtual Machines & Kubernetes:

- Virtual machines host the application services, while managed Kubernetes automatically deploys, scales, and manages application containers.
- This ensures high performance, reliability, and fault tolerance during peak order times.

Real-Time Integration:

- The system integrates clients, APIs, cloud storage, and cloud infrastructure in real time.
- Customers receive instant updates on order status, while restaurants receive new orders immediately.

Advantages:

- Fast and scalable food ordering system.
- Secure storage of customer and order data.
- High availability through cloud infrastructure.
- Automatic scaling using Kubernetes.
- Real-time order processing and tracking.